

Aditya Saini

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EDUCATION

Purdue University

West Lafayette, IN

B.S. Computer Science (Systems Programming)

Graduation: May 2028 (Expected)

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Mathematics, Computer Architecture, Programming in C, Object-Oriented Programming, Multivariate Calculus

EXPERIENCE

Embedded Engineer

Sep. 2025 – Present

Purdue Electric Racing

West Lafayette, IN

- Designed, developed, tested, and debugged modular RTOS-based vehicle firmware in C, integrating drivers with hardware via in-circuit debugging
- Implemented runtime telemetry and data acquisition to profile and validate performance bottlenecks in production firmware
- Designed I2C and SPI HALs for STM32 platforms from component datasheets, building reusable infrastructure APIs for high-frequency sensor communication
- Developed Nextion LCD UART drivers and optimized command encoding for reliable real-time dashboard updates

Chip Design Member (SoCET STARS)

June 2026 – August 2026

SoCET, Purdue VIP

West Lafayette, IN

- Developed memory-mapped I/O (MMIO) peripheral drivers and accelerator integration for a custom RISC-V SoC, deepening computer-architecture and peripheral fundamentals
- Designed combinational and sequential logic (decoders, registers, FSMs) in SystemVerilog RTL and validated each block with self-checking testbenches through the RTL-to-GDSII flow
- Wrote RISC-V assembly for the AFTx07 ISA, read schematics/datasheets, and managed builds with Embedded Linux, Makefiles, and Git on the asicfab server

Robotic Safety Research Assistant

January 2026 – July 2026

PurSec, Purdue Computer Science

West Lafayette, IN

- Researched jailbreak vulnerabilities in multimodal-LLM-controlled robots, uncovering multi-step attacks that chain individually-safe actions to defeat state-of-the-art safety systems
- Built automated Python test and evaluation pipelines on headless Linux with Docker and AWS EC2, delivering reproducible regression runs and metrics across 10+ scenarios

PROJECTS

Natural Language Shell | C, Python, OpenAI API, Git

June 2025 – January 2026

- Created a lightweight command-line shell in C supporting natural language commands with tokenizer, intent parsing, and bash mapping; integrated OpenAI API for complex commands/queries through OpenRouter.

FRC Team 2554 | Java, OpenCV, Jetson Orin Nano, CUDA

September 2021 – August 2025

- Led the team to its best-ever season as Build President, bringing us to District Championships after 18+ years.
- Developed and led our team's first instance of Swerve Drive, tuned PID/FF loops, and developed auto commands to increase accuracy by 66% using custom vision pipelines to detect AprilTags for robot positioning.

Tree Generator | Rust, CLI, Data Structures

April 2026

- Built a Rust CLI that generates configurable general and binary trees with visual printing and DFS traversals (preorder, inorder, postorder), applying ownership-safe data structures and recursion.

TECHNICAL SKILLS

Languages: C, C++, Assembly, Python, Rust, SystemVerilog, Java, SQL

Embedded & Firmware: RTOS, bare-metal firmware, STM32, ESP32, device drivers, HALs, SPI, I2C, UART, CAN, in-circuit debugging, board bring-up

Systems & Tools: Embedded Linux, Git, Makefiles, GDB, Docker, Unit/Integration testing, datasheets & schematics

CS Foundations: Data Structures & Algorithms, Object-Oriented Design, Computer Architecture, Operating Systems, Concurrency, Networking (TCP/IP)